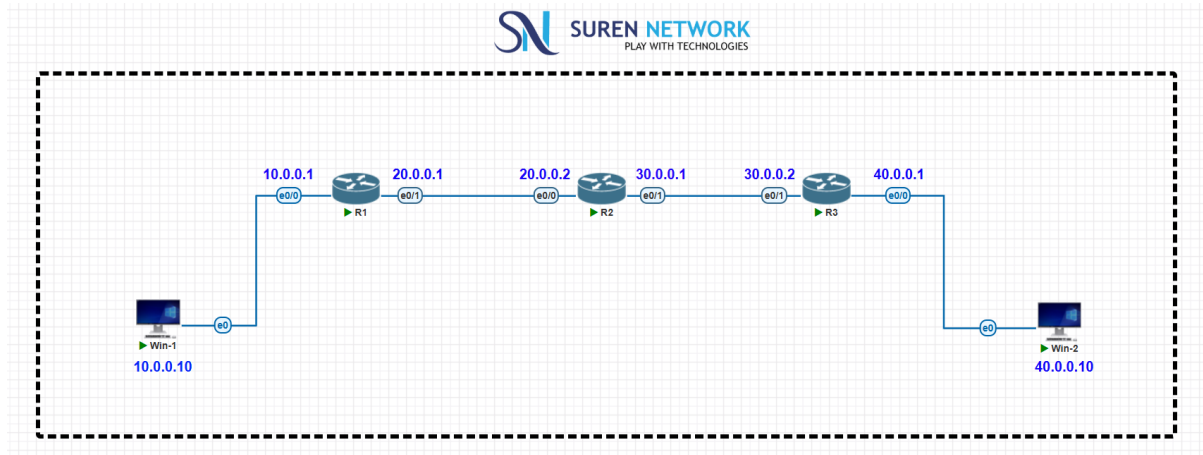


## OSPF



```
R1
Ethernet0/2          unassigned      YES TFTP   down      down
Ethernet0/3          unassigned      YES TFTP   down      down

R1#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#router-id 1.1.1.1
R1(config-router)#network 10.0.0.0 0.255.255.255 area 1
R1(config-router)#network 20.0.0.0 0.255.255.255 area 1
R1(config-router)#exit
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#
*Sep 27 14:32:32.698: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(config)#
*Sep 27 14:33:57.592: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet0/1 from LOADING to FULL, Loading Done
R1(config)#
```

```
R2
Ethernet0/1          30.0.0.1      YES TFTP   up        up
Ethernet0/2          unassigned  YES TFTP   down      down
Ethernet0/3          unassigned  YES TFTP   down      down

R2#
R2#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
*Sep 27 14:32:38.583: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R2(config)#router ospf 2
R2(config-router)#router-id 2.2.2.2
R2(config-router)#network 20.0.0.0 0.255.255.255 area 1
R2(config-router)#network 20.0.0.0 0.255.255.255 area 1
*Sep 27 14:33:57.593: %OSPF-5-ADJCHG: Process 2, Nbr 1.1.1.1 on Ethernet0/0 from LOADING to FULL, Loading Done
R2(config-router)#network 30.0.0.0 0.255.255.255 area 1
R2(config-router)#exit
R2(config)#
*Sep 27 14:36:23.489: %OSPF-5-ADJCHG: Process 2, Nbr 3.3.3.3 on Ethernet0/1 from LOADING to FULL, Loading Done
R2(config)#
```

```
R3
Ethernet0/1          30.0.0.2      YES TFTP   up        up
Ethernet0/2          unassigned  YES TFTP   down      down
Ethernet0/3          unassigned  YES TFTP   down      down

R3#
R3#
R3#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router ospf 3
R3(config-router)#router-id 3.3.3.3
R3(config-router)#network 30.0.0.0 0.255.255.255 area 1
R3(config-router)#network 30.0.0.0 0.255.255.255 area 1
*Sep 27 14:36:23.489: %OSPF-5-ADJCHG: Process 3, Nbr 2.2.2.2 on Ethernet0/1 from LOADING to FULL, Loading Done
R3(config-router)#network 40.0.0.0 0.255.255.255 area 1
R3(config-router)#exit
R3(config)#
*Sep 27 14:39:18.992: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R3(config)#
```

```
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
2.2.2.2          1     FULL/BDR        00:00:35    20.0.0.2     Ethernet0/1
R1#
R1#
R1#
R1#
R1#
R1#
```

```
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#show ip ospf database

        OSPF Router with ID (1.1.1.1) (Process ID 1)

        Router Link States (Area 1)

Link ID        ADV Router    Age          Seq#           Checksum Link count
1.1.1.1        1.1.1.1      833          0x80000005    0x0017B8 2
2.2.2.2        2.2.2.2      688          0x80000006    0x003959 2
3.3.3.3        3.3.3.3      643          0x80000005    0x008606 2

        Net Link States (Area 1)

Link ID        ADV Router    Age          Seq#           Checksum
20.0.0.1       1.1.1.1      833          0x80000001    0x00F81C
30.0.0.1       2.2.2.2      688          0x80000001    0x00AC52
R1#
```

```

R1#
R1#
R1#show ip ospf interface eth0/0
Ethernet0/0 is up, line protocol is up
  Internet Address 10.0.0.1/8, Area 1, Attached via Network Statement
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 10
  Topology-MTID      Cost      Disabled      Shutdown      Topology Name
    0                10         no           no           Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 10.0.0.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:07
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1/1, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 0
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
R1#

```

```

R1#
R1#
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
O       30.0.0.0/8 [110/20] via 20.0.0.2, 00:17:05, Ethernet0/1
O       40.0.0.0/8 [110/30] via 20.0.0.2, 00:14:44, Ethernet0/1
R1#

```

